Networking concepts and security

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Unit - 3

Proxy & VPNs

VPN

A Virtual Private Network (VPN) is a technology that allows users to create a secure and encrypted connection to a private network over the internet. VPNs are commonly used to protect online privacy, secure data transmission, and access restricted content.

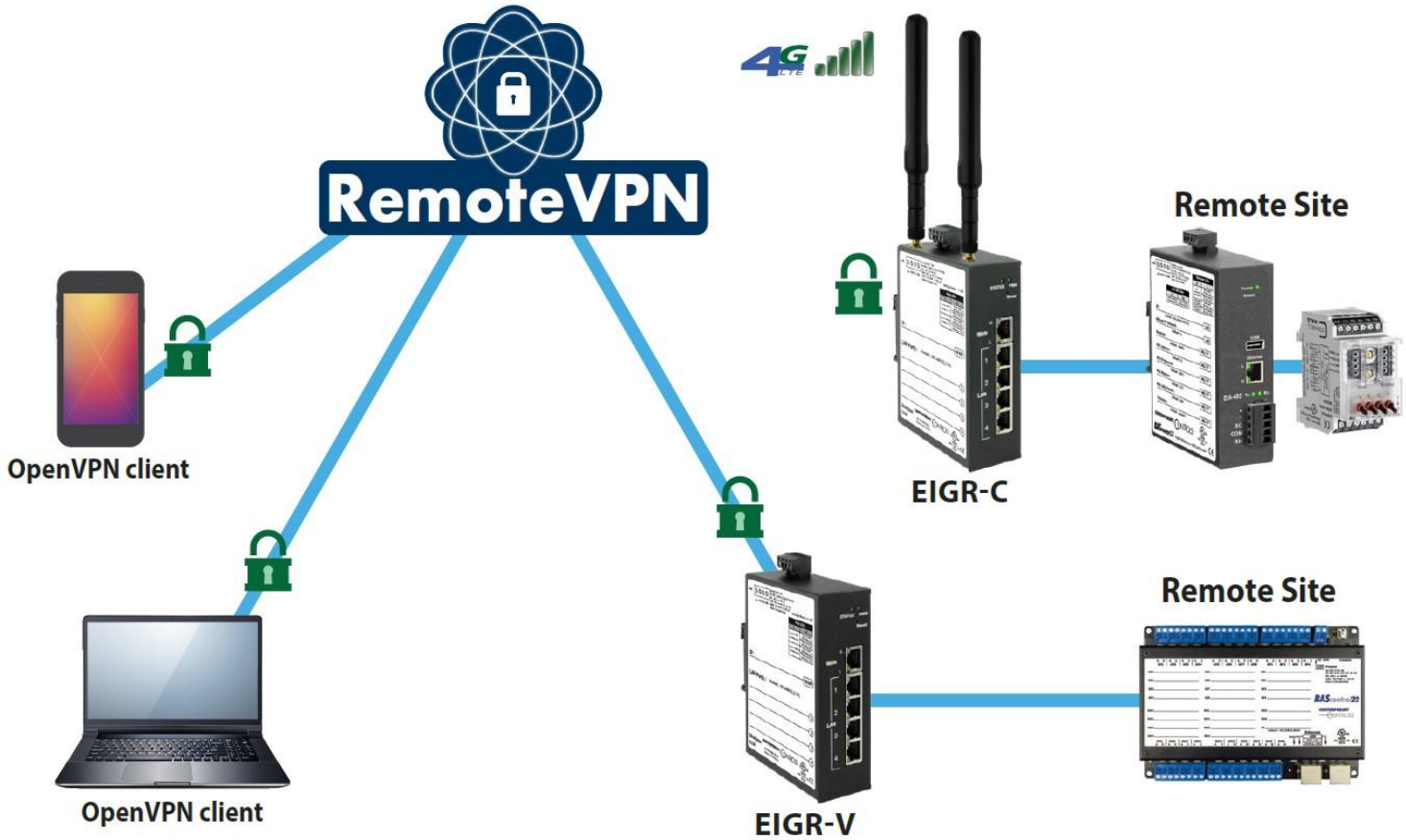
VPN
Secures your Data
VPN
YOU
THE INTERNET
GOVERNMENT
HACKERS
SNOOPERS

VPN types

Remote Access VPN:

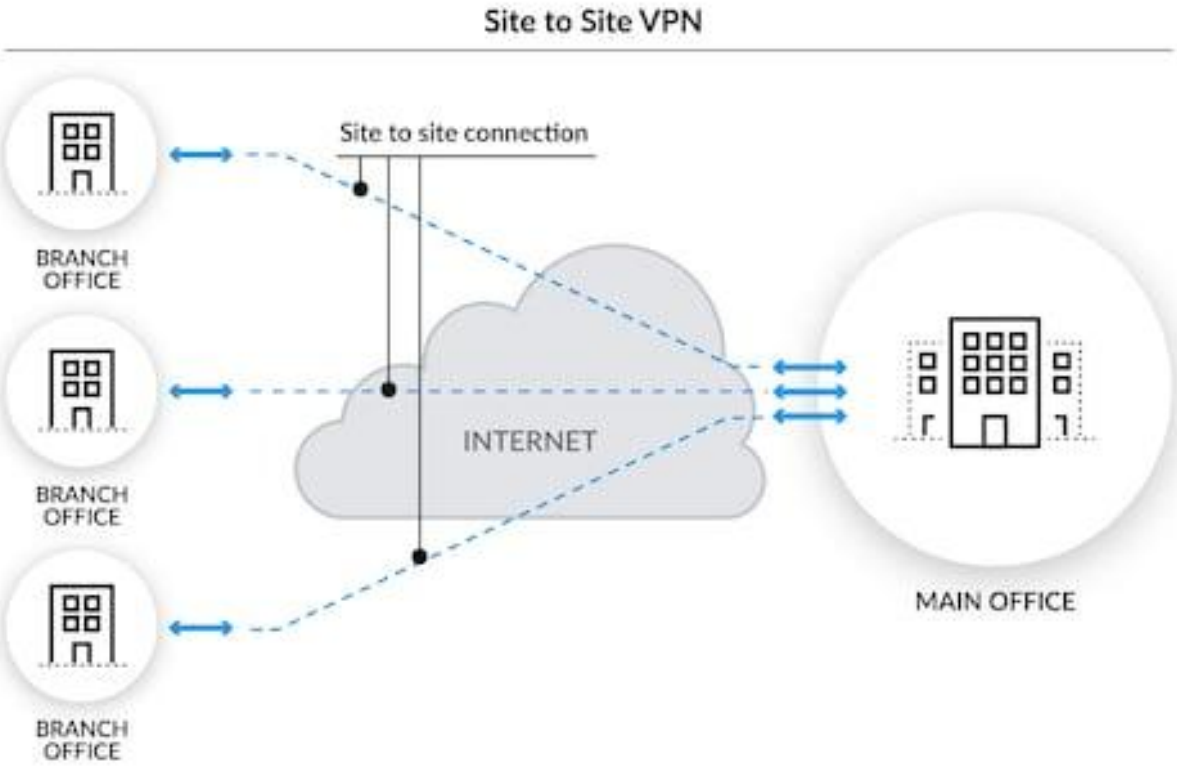
Purpose: Provides a secure connection for remote users to access a private network over the internet.

Use Case: Employees working from home or traveling can connect to their company's internal network securely.


RemoteVPN
4G
Remote Site
OpenVPN client
EIGR-C
Remote Site
OpenVPN client
EIGR-V

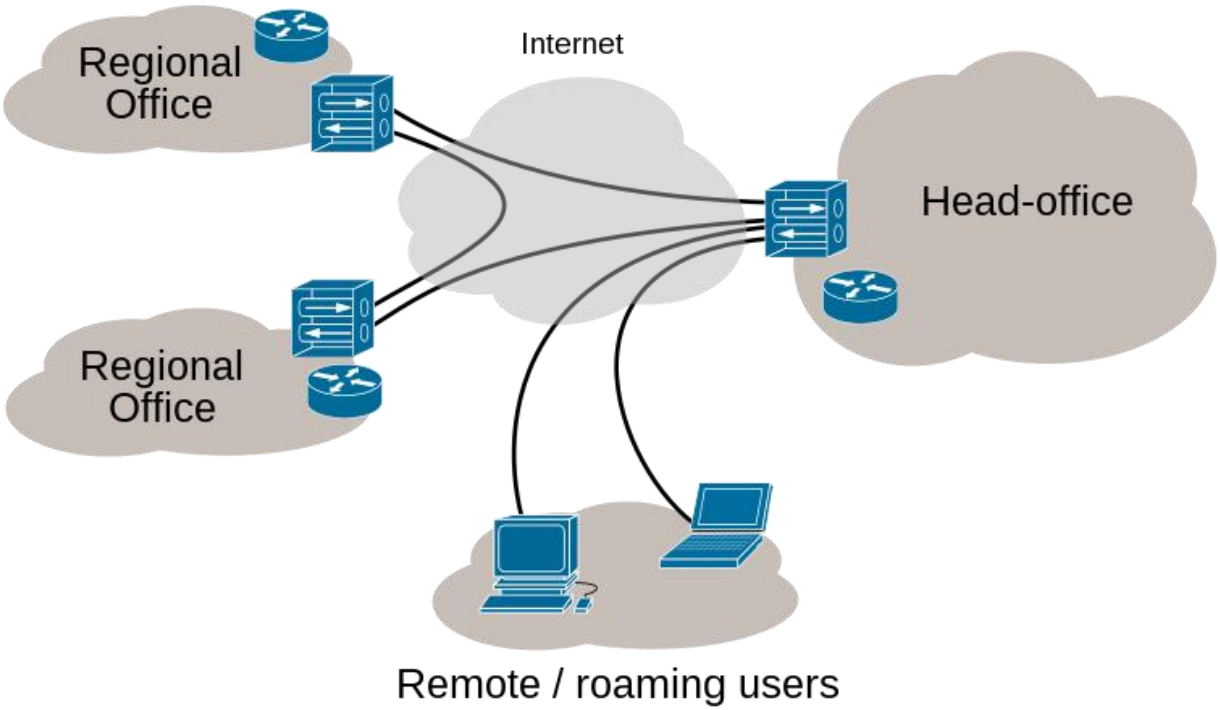
VPN types

Site-to-Site VPN:
Purpose: Connects entire networks together, allowing multiple offices or branches of a company to share data securely.
Use Case: Companies with multiple locations can use site-to-site VPNs to connect their local networks securely over the internet.

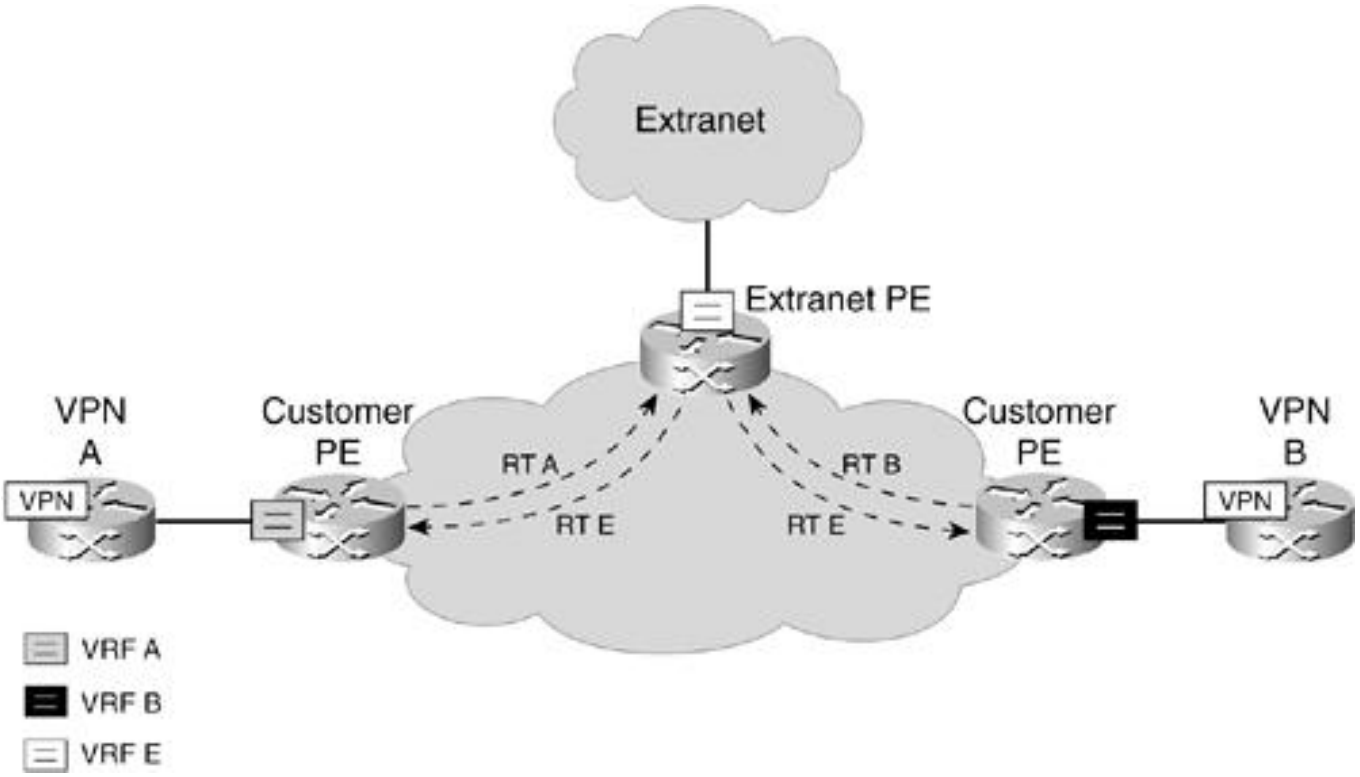

Site to Site VPN
Site to site connection
BRANCH OFFICE
BRANCH OFFICE
BRANCH OFFICE
INTERNET
MAIN OFFICE

Intranet-based VPN:
Purpose: Connects different branches or remote offices of the same organization to create a secure communication network.
Use Case: Large organizations with geographically dispersed offices can use intranet-based VPNs for secure internal communication.

Internet VPN



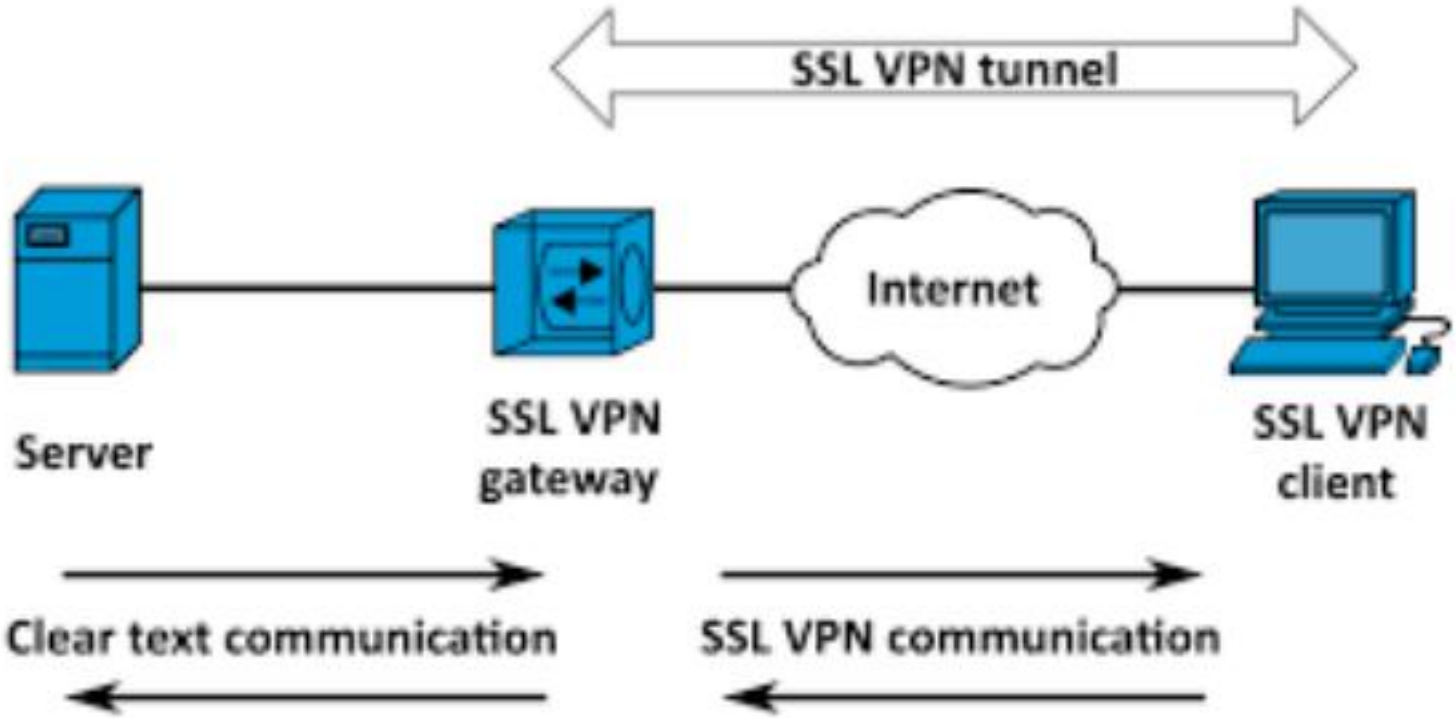
Extranet-based VPN:
Purpose: Allows different organizations to establish a secure communication channel over the internet.
Use Case: Partner companies or organizations can share data securely while maintaining privacy and security.


Extranet
Extranet PE
VPN
A
VPN
Customer
PE
RT A
RT E
RT B
RT E
Customer
PE
VPN
B
VPN
VRF A
VRF B
VRF E

SSL/TLS VPN:

Purpose: Uses SSL/TLS protocols to provide a secure connection over the web browser, often used for secure remote access.

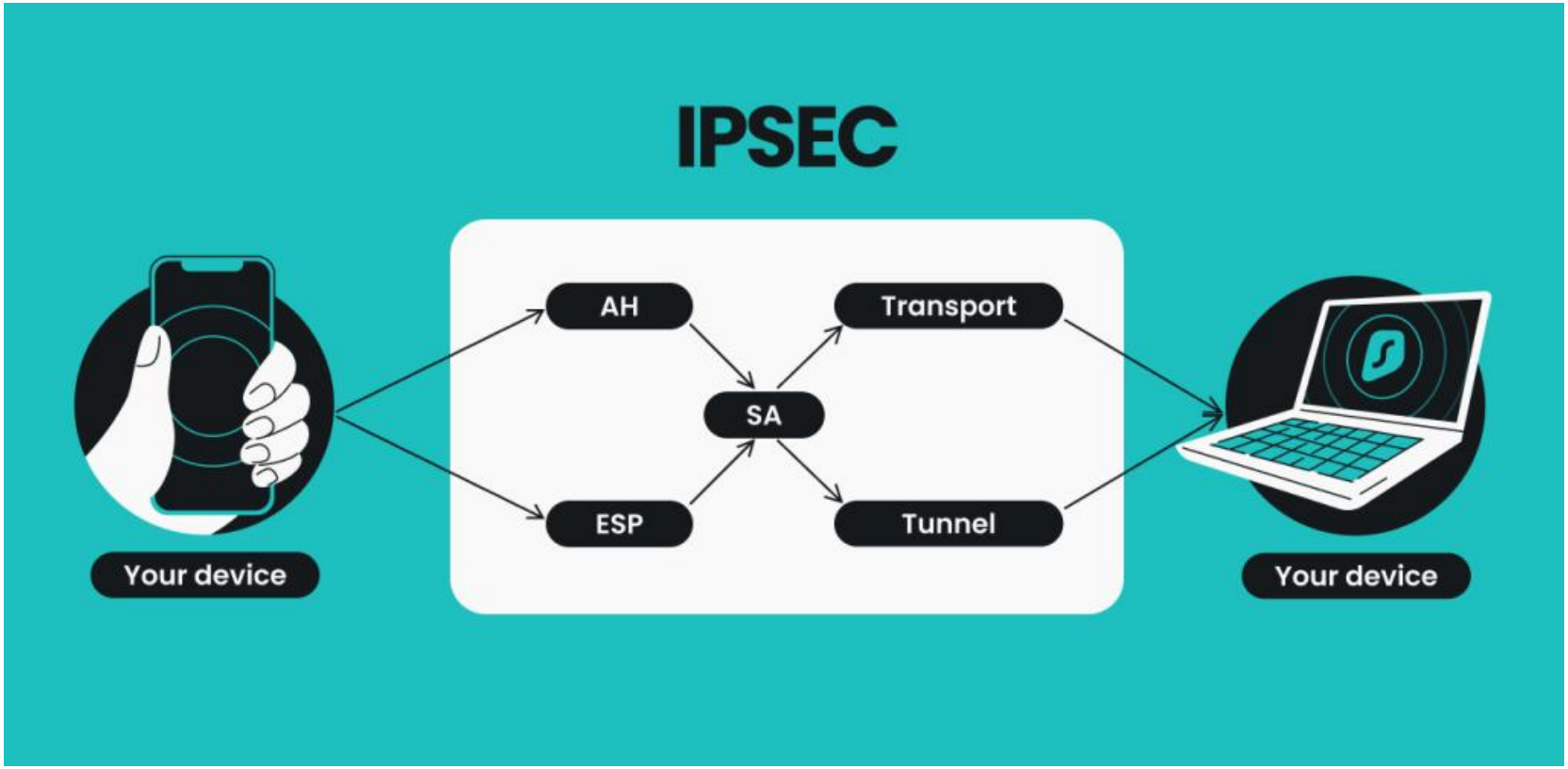
Use Case: Securely accessing corporate applications or data through a web browser without requiring additional software.


SSL VPN tunnel
Internet
Server
SSL VPN
gateway
SSL VPN
client
Clear text communication
SSL VPN communication

IPsec VPN:

Purpose: Uses Internet Protocol Security (IPsec) to provide a secure connection between networks or devices.

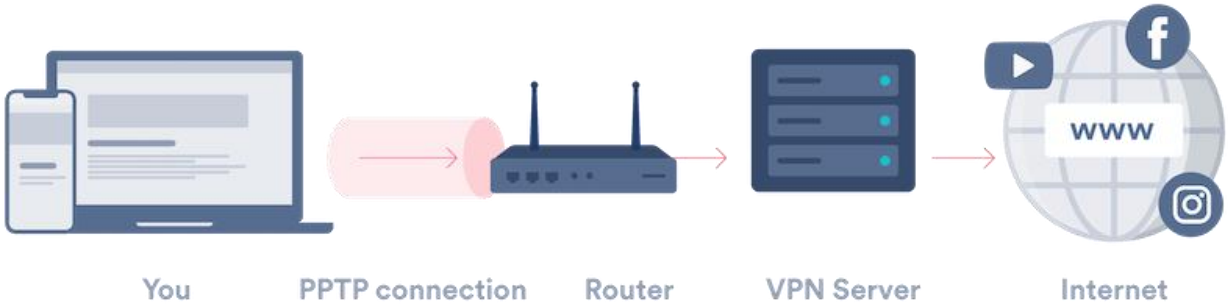
Use Case: Commonly used for site-to-site VPNs and remote access VPNs, offering a high level of security.


IPSEC
AH
ESP
SA
Transport
Tunnel
Your device
Your device

PPTP VPN:

Purpose: Point-to-Point Tunneling Protocol (PPTP) was one of the earliest VPN protocols but is now considered less secure due to vulnerabilities.

Use Case: Legacy systems that still support PPTP, though it's not recommended for use due to security concerns.

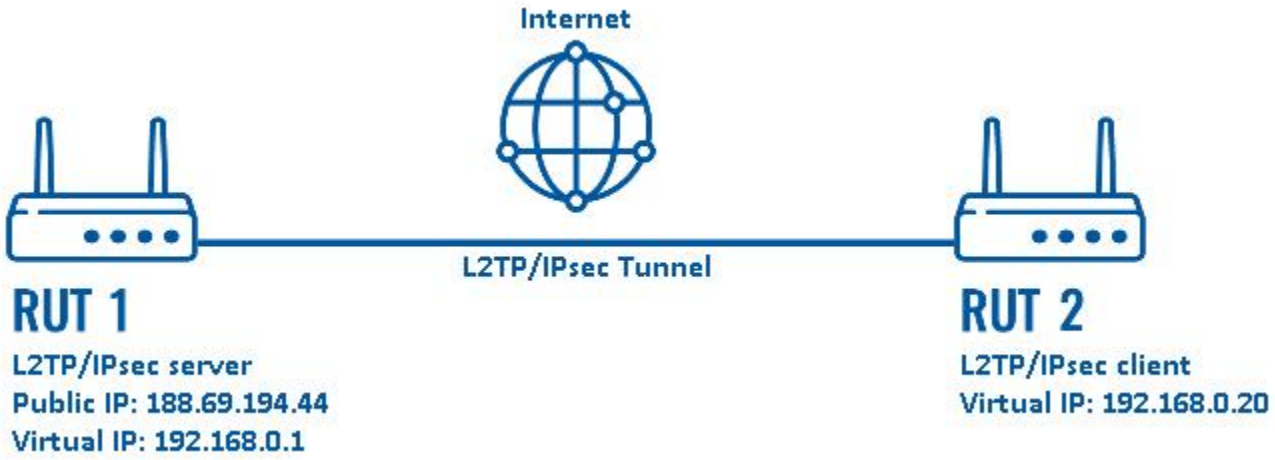


Surfshark®

L2TP/IPsec VPN:

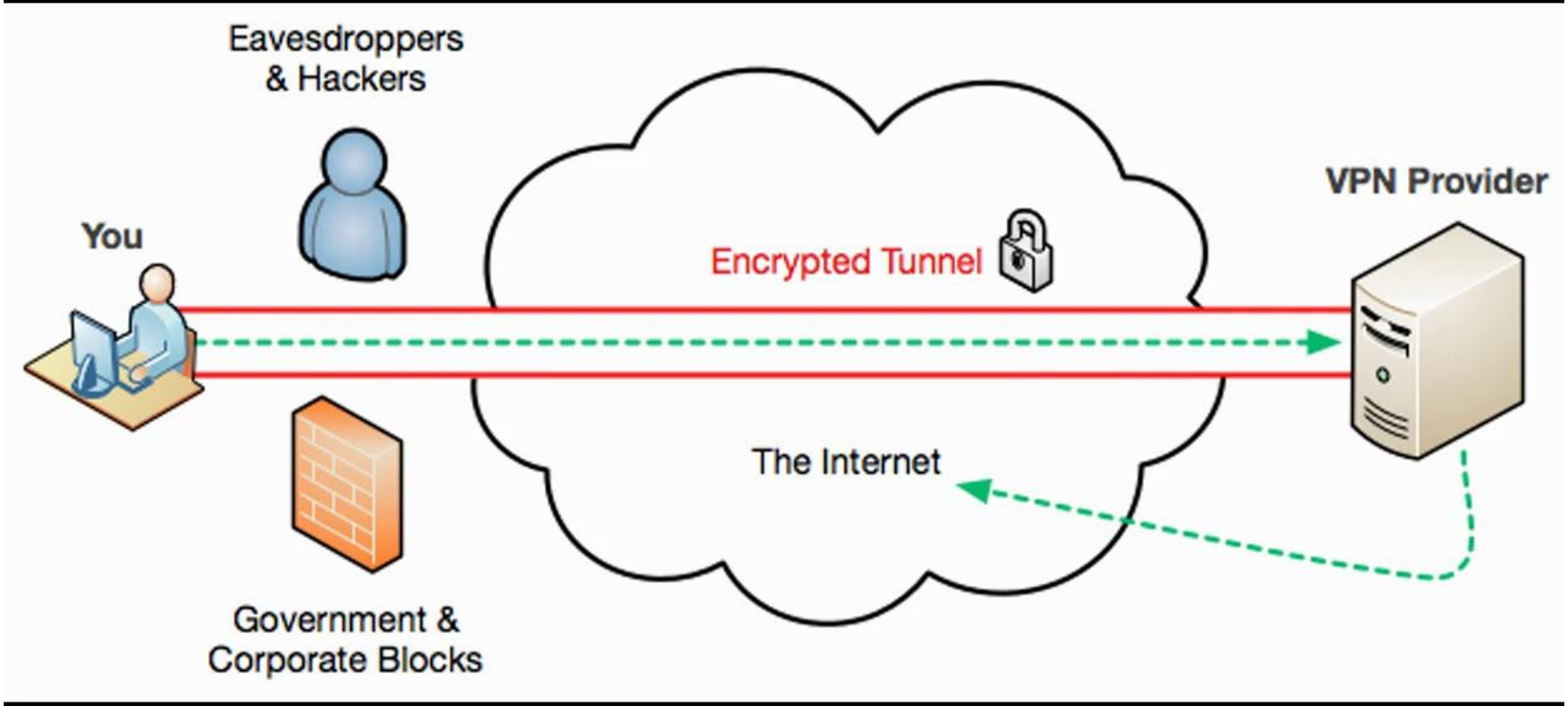
Purpose: Layer 2 Tunneling Protocol (L2TP) combined with IPsec to provide a more secure and robust VPN connection.

Use Case: Offers a higher level of security than PPTP and is commonly used for remote access VPNs.


Internet
L2TP/IPsec Tunnel
RUT 1
L2TP/IPsec server
Public IP: 188.69.194.44
Virtual IP: 192.168.0.1
RUT 2
L2TP/IPsec client
Virtual IP: 192.168.0.20

Tunnelling Protocols

Tunneling protocols are a crucial aspect of Virtual Private Networks (VPNs) as they are responsible for encapsulating and encrypting data for secure transmission over the internet.


Eavesdroppers
& Hackers
You
Encrypted Tunnel
VPN Provider
The Internet
Government &
Corporate Blocks

Tunneling protocols:

Point-to-Point Tunneling Protocol (PPTP):

Purpose: Provides a basic level of encryption and is easy to set up.

Security: Considered less secure due to vulnerabilities and is not recommended for sensitive data.

Tunneling protocols:

Layer 2 Tunneling Protocol (L2TP):

Purpose: Often used in combination with IPsec for improved security.

Security: L2TP alone doesn't provide encryption, but when used with IPsec, it becomes more secure.

Tunneling protocols:

Internet Protocol Security (IPsec):

Purpose: Adds a layer of security by authenticating and encrypting each IP packet.

Security: Widely used for site-to-site VPNs and remote access VPNs due to its strong security features.

Secure Socket Tunneling Protocol (SSTP):
Purpose: Developed by Microsoft and operates over the SSL/TLS protocol.
Security: Offers a high level of security and is commonly used for remote access VPNs on Windows platforms.

OpenVPN:
Purpose: An open-source protocol that uses SSL/TLS for key exchange and offers flexibility and security.
Security: Considered highly secure and is widely used for both remote access and site-to-site VPNs.

WireGuard:

Purpose: A modern, open-source VPN protocol designed for simplicity and efficiency.

Security: Known for its speed and security, with a smaller codebase compared to some other protocols.

IKEv2/IPsec:

Purpose: A combination of Internet Key Exchange version 2 (IKEv2) and IPsec for strong security and quick reconnection after a network interruption.

Security: Provides robust security and is suitable for mobile devices due to its ability to handle changes in network connectivity seamlessly.

Tunnel and Transport Mode

In the context of Virtual Private Networks (VPNs) and IPsec (Internet Protocol Security), "Tunnel mode" and "Transport mode" refer to two different ways in which IPsec can be applied to secure communication between two devices or networks.

Tunnel Mode:

Purpose: Tunnel mode is typically used to create VPNs between networks.

Encapsulation: In tunnel mode, the entire original IP packet (including the original IP header) is encapsulated within a new IP packet.

Tunnel Mode:

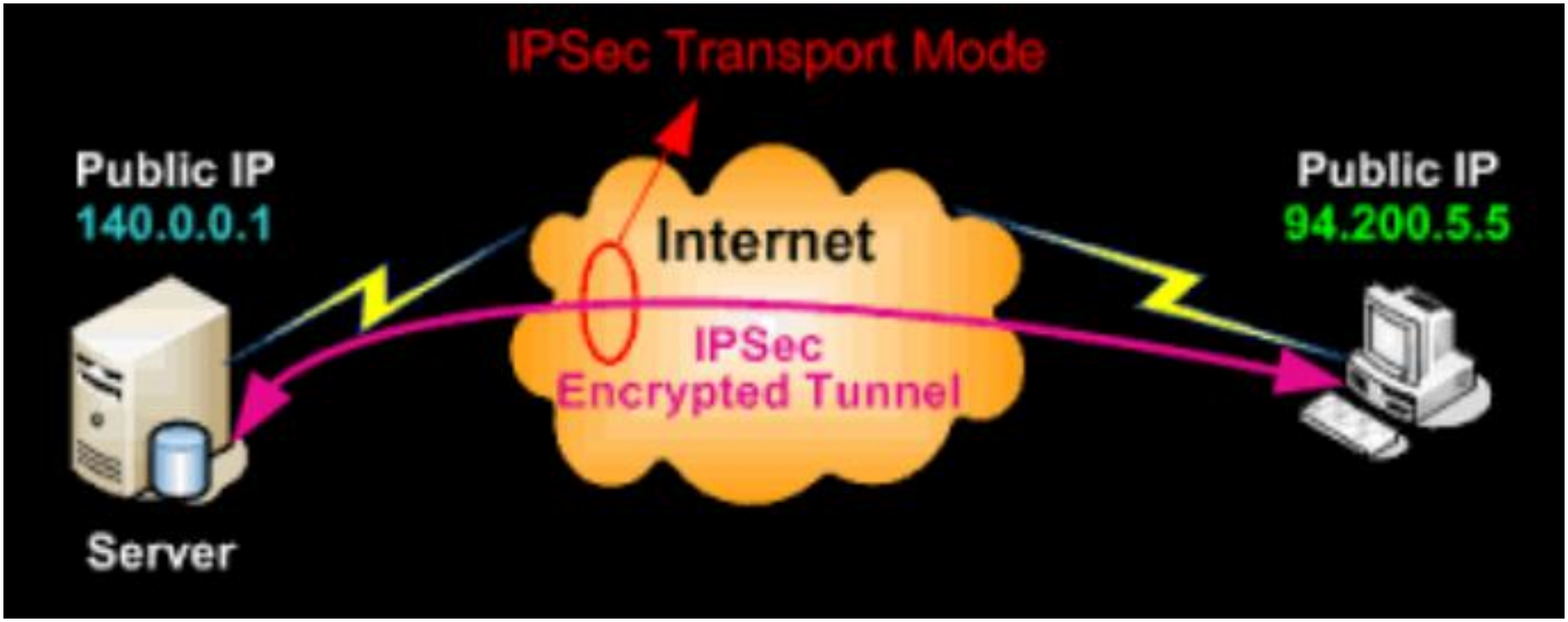
Header Modification: The original IP header is usually modified, with the new header containing information related to the secure communication between the two endpoints.

Use Case: Commonly used in site-to-site VPNs, where entire networks are connected securely over the internet.

Transport Mode:

Purpose: Transport mode is typically used for end-to-end communication between two devices.

Encapsulation: In transport mode, only the payload (data) of the original IP packet is encrypted and/or authenticated, leaving the original IP header intact.


IPSec Transport Mode
Public IP
140.0.0.1
Internet
Public IP
94.200.5.5
IPSec
Encrypted Tunnel
Server

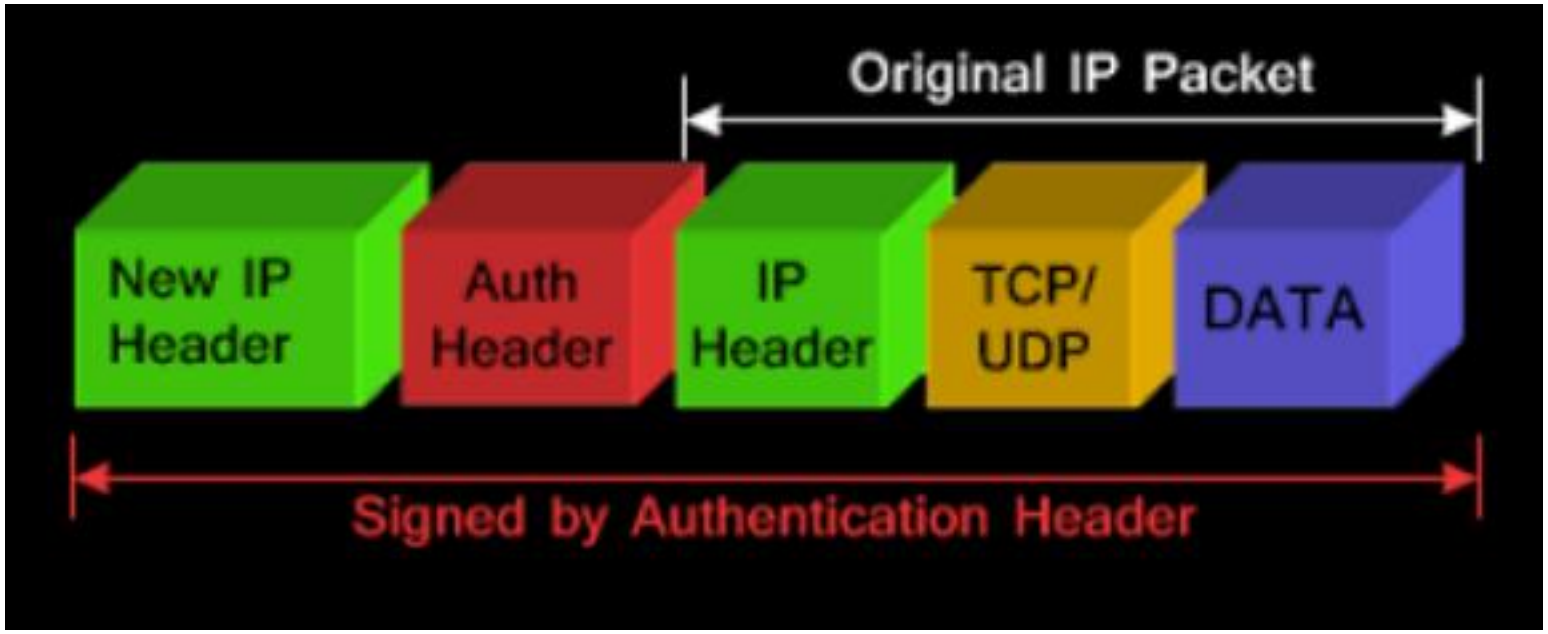
Transport Mode:

Header Modification: The original IP header is not modified; instead, additional headers are added to the packet to provide security services.

Use Case: Often used for securing communication between individual devices, such as in remote access VPNs or point-to-point communication.

Authentication Header

The Authentication Header (AH) is one of the two main protocols used in IPsec (Internet Protocol Security) to provide security services for IP packets. The other protocol is the Encapsulating Security Payload (ESP). Both AH and ESP can be used in conjunction, but they serve different purposes.


Original IP Packet
New IP Header
Auth Header
IP Header
TCP/ UDP
DATA
Signed by Authentication Header

Purpose of Authentication Header (AH):

Authentication:

Integrity: AH provides data integrity by ensuring that the contents of the IP packet have not been altered in transit. It achieves this by calculating a cryptographic hash of the packet's contents and including it in the AH header.

Authentication: This cryptographic hash, also known as the Integrity Check Value (ICV), is used to verify the authenticity of the packet. If the packet has been tampered with during transmission, the hash will not match, and the packet will be rejected.

Purpose of Authentication Header (AH):

Anti-Replay Protection:

AH includes a sequence number field to protect against replay attacks. This field is used to order incoming packets and identify if any packets are duplicates or out of order.

No Confidentiality:
Unlike ESP, AH does not provide encryption. It only focuses on ensuring the integrity and authenticity of the transmitted data.

Structure of Authentication Header:

The AH is inserted between the original IP header and the upper-layer protocol (e.g., TCP or UDP). The structure of the Authentication Header includes several fields, but the most important ones are:

Next Header: Identifies the type of the next header, indicating the upper-layer protocol (e.g., TCP, UDP) following the AH header.

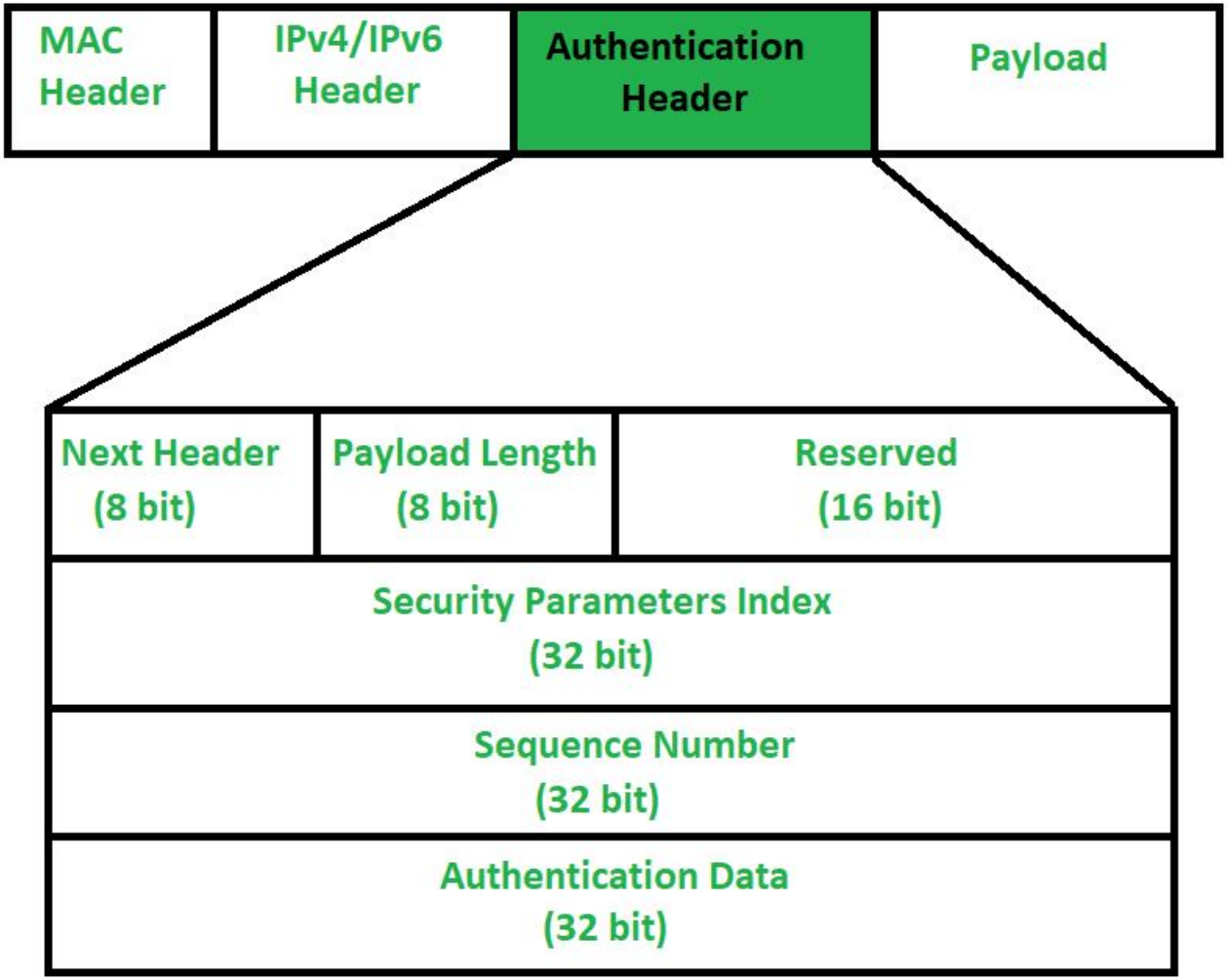
Payload Length: Specifies the length of the AH header and the payload it covers.

Reserved: Reserved for future use.

Security Parameters Index (SPI): A unique identifier for a specific security association.

Sequence Number: Prevents replay attacks by assigning a unique number to each packet.

Authentication Data (ICV): The Integrity Check Value, containing the cryptographic hash used for data integrity and authentication.


MAC Header
IPv4/IPv6 Header
Authentication Header
Payload
Next Header (8 bit)
Payload Length (8 bit)
Reserved (16 bit)
Security Parameters Index (32 bit)
Sequence Number (32 bit)
Authentication Data (32 bit)

IPSEC Protocol Suite

The IPsec (Internet Protocol Security) protocol suite is a comprehensive set of protocols and standards designed to secure Internet Protocol (IP) communications. IPsec provides a range of security services such as authentication, integrity, confidentiality, and replay protection for IP packets. It is widely used to establish Virtual Private Networks (VPNs) and to secure communication over IP networks. The IPsec suite consists of several protocols, with the two main components being the Authentication Header (AH) and the Encapsulating Security Payload (ESP). Additionally, key management protocols are used to establish and manage cryptographic keys for secure communication.

Components:

Authentication Header (AH):
Purpose: Provides authentication and integrity services for IP packets.

Authentication: Verifies the authenticity of the transmitted data.

Integrity: Ensures that the contents of the IP packet have not been altered during transmission.

Anti-Replay Protection: Guards against replay attacks using sequence numbers.

Encapsulating Security Payload (ESP):
Purpose: Provides confidentiality, authentication, and integrity services for IP packets.

Confidentiality: Encrypts the payload of the IP packet to keep it confidential.

Authentication: Similar to AH, ESP also provides authentication and integrity services.

Optional: Anti-Replay Protection: ESP can include measures for protection against replay attacks.

Security Associations (SA):
Purpose: Defines a one-way logical connection between two entities (such as two devices or networks) for secure communication.

Attributes: Each SA has specific attributes, including the security protocol (AH or ESP), cryptographic algorithms, and keys.

Establishment: SAs are established through a process known as negotiation, typically involving the Internet Key Exchange (IKE) protocol.

Internet Key Exchange (IKE):
Purpose: Used to automate the negotiation and management of security associations (SAs) between devices.

Authentication: Authenticated key exchange protocols are used to establish a secure communication channel.

Key Management: Handles the exchange of cryptographic keys for secure communication.
Phase 1 and Phase 2: IKE negotiation consists of two phases, with Phase 1 establishing a secure channel for further negotiation in Phase 2.

Key Management:
Purpose: Involves the generation, distribution, and management of cryptographic keys used by IPsec.

Manual or Automated: Keys can be configured manually or established through automated protocols like IKE.

Lifetime Management: Defines how long a set of cryptographic keys is valid before being refreshed or renegotiated.

Tunnel Mode and Transport Mode:
Tunnel Mode: Used for VPNs, where entire IP packets, including the original IP header, are encapsulated and encrypted for secure transmission.

Transport Mode: Used for end-to-end communication, where only the payload of the IP packet is encrypted, leaving the original IP header intact.

IKE PHASE 1

Internet Key Exchange (IKE) is a key management protocol that is a part of the IPsec (Internet Protocol Security) suite. IKE is used to establish a secure and authenticated communication channel by negotiating and exchanging cryptographic keys between two devices, such as VPN gateways or hosts. IKE operates in two phases: Phase 1 and Phase 2.

IKE Phase 1

Objective: Establish a secure, authenticated communication channel and negotiate a shared secret key for further use in securing communication (Phase 2).

Key Aspects:

Authentication:

Purpose: Ensure the identities of the communicating parties and prevent unauthorized access.

IKE Phase 1

Methods: IKE Phase 1 supports various authentication methods, such as pre-shared keys (PSK), digital certificates, or a combination of both.

Encryption Algorithm Negotiation:
Purpose: Determine the encryption algorithm to be used for protecting the confidentiality of the IKE communication.
Options: Negotiation includes selecting an encryption algorithm, such as DES, 3DES, AES, etc.

Lifetime Negotiation:
Purpose: Define the duration for which the negotiated keys and security associations will be valid before re-negotiation is required.
Lifetime Parameters: Typically expressed in seconds and/or kilobytes.

Main Mode or Aggressive Mode:
Main Mode: A three-message negotiation process that is more secure but requires more messages.

Aggressive Mode: A quicker negotiation process using fewer messages but with slightly less security due to some information being exchanged in the clear.

Lifetime Negotiation:

Purpose: Define the duration for which the negotiated keys and security associations will be valid before re-negotiation is required.

Lifetime Parameters: Typically expressed in seconds and/or kilobytes.

IKE Phase 1 Process:

Main Mode or Aggressive Mode:
Main Mode: A three-message negotiation process that is more secure but requires more messages.

Aggressive Mode: A quicker negotiation process using fewer messages but with slightly less security due to some information being exchanged in the clear.

Authentication:
The parties authenticate each other using the agreed-upon method (e.g., pre-shared keys, digital certificates).

Establishment of IKE SA (Security Association):
A secure IKE SA is established, and the negotiated parameters are used to derive the keying material for further use in Phase 2.

Implementation of VPNs

The implementation of Virtual Private Networks (VPNs) involves setting up a secure and encrypted communication channel over an existing network, typically the internet. There are several methods and technologies for implementing VPNs, each with its own advantages and use cases.

VPN implementations:

Remote Access VPNs:

Purpose: Provides secure access to a private network for individual users or remote devices.

VPN implementations:

Implementation:
VPN Client Software: Users install VPN client software on their devices.
Authentication: Users typically authenticate using credentials, such as a username and password.

Encryption: The client establishes a secure connection to a VPN server using protocols like PPTP, L2TP/IPsec, IKEv2/IPsec, or SSL/TLS.
Examples: SSL VPNs (web-based VPNs), IPsec-based client VPNs.

Site-to-Site VPNs:

Purpose: Connects multiple networks or sites securely, allowing communication as if they were on the same physical network.

Implementation:
VPN Gateway/Firewall: Each site has a VPN gateway or firewall that manages the secure connection.

Authentication: VPN gateways authenticate each other using pre-shared keys or digital certificates.

Encryption: IPsec is commonly used for encryption and tunneling between sites.
Examples: IPsec-based VPNs, MPLS (Multiprotocol Label Switching) VPNs.

Encryption: SSL/TLS protocols secure the communication between the client and the VPN server.
Examples: Cisco AnyConnect, OpenVPN with SSL/TLS, and other SSL-based VPN solutions.

Clientless VPNs:

Purpose: Allows secure access to resources without requiring the installation of client software.

Implementation:
Web Browser Interface: Users connect to a web portal using a standard web browser.

Authentication: Users log in using credentials through the web interface.

Access: Users can access resources through a web-based interface.
Examples: Cisco SSL VPN, Juniper Networks Secure Access.

OpenVPN:

Purpose: Open-source VPN solution providing flexibility and security.

Implementation:
OpenVPN Software: OpenVPN software is installed on both the client and server.

Authentication: Various methods, including username/password or certificates.

Encryption: Uses SSL/TLS protocols for secure communication.

Examples: OpenVPN is commonly used in various configurations, including site-to-site and remote access VPNs.

WireGuard:

Purpose: Modern and lightweight open-source VPN protocol.

Implementation:
WireGuard Software: WireGuard software is used on client and server devices.

Authentication: Public and private keys are used for authentication.

Encryption: Utilizes modern cryptography for secure communication.

Examples: Growing in popularity due to its simplicity and performance.

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